

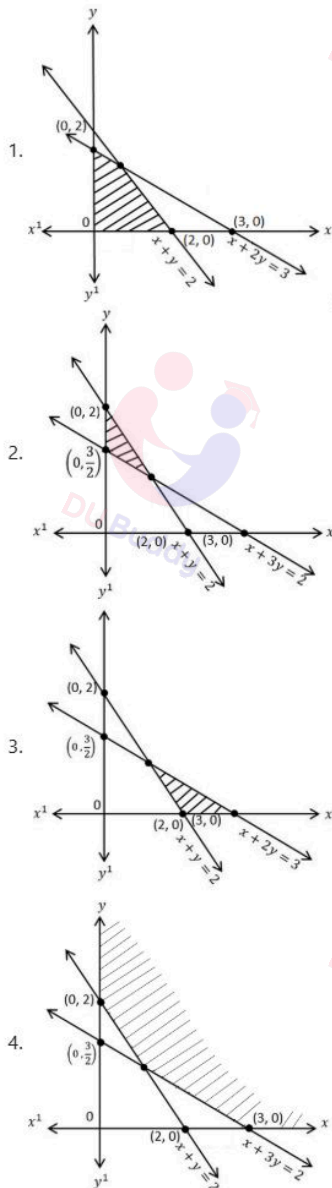
Section : Common

Q.1

For LPP: Maximize $z = 2x + 3y$ subject to the constraints

$$x + y \geq 2, x + 2y \geq 3, x \geq 0, y \geq 0,$$

which of the following graph represents the feasible region of the above LPP as shaded portion?



Options 1. 1

2. 2

3. 3

4. 4

Q.2

If $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 5 & -2 \end{pmatrix}$, be such that $\mathbf{A}^{-1} = \mathbf{K}\mathbf{A}$, then the value of $\mathbf{K}$ is:

1. 19

2. $\frac{1}{19}$

3. 0

4. -19

Options 1. 1

2. 2

3. 3
4. 4

Q.3

The maximum value of $\left(\frac{1}{x}\right)^x$ for $x > 0$ is

1. e
2. $e^{1/e}$
3. $\left(\frac{1}{e}\right)^{1/e}$
4. e^e

Options

1. 1
2. 2
3. 3
4. 4

Q.4

The solution of the differential equation $\frac{dy}{dx} = (1+x^2)(1+y^2)$ is (Here C is an arbitrary constant)

1. $\tan^{-1} y + x + \frac{x^3}{3} = C$
2. $\tan^{-1} y - x - \frac{x^3}{3} = C$
3. $\tan^{-1} x - y - \frac{y^3}{3} = C$
4. $\tan^{-1} x + y + \frac{y^3}{3} = C$

Options

1. 1
2. 2
3. 3
4. 4

Q.5

The largest interval, in which the function $f(x) = x^3 + 2x^2 - 1$ is increasing, is:

1. $(0, \infty)$
2. $(-4, 4)$
3. $\left[-\frac{4}{3}, 0\right]$
4. $(-\infty, -\frac{4}{3}] \cup [0, \infty)$

Options

1. 1
2. 2
3. 3
4. 4

Q.6

Let $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -6 \\ -2 & 4 \end{bmatrix}$, then

- (A) $\det(A^T) = 1$
(B) $AB = I$, where I is the identity matrix of order 2
(C) $A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$
(D) $\text{adj}(B) = \begin{bmatrix} 4 & 2 \\ 6 & 4 \end{bmatrix}$

Choose the correct answer from the options given below:

1. (A) and (C) only

2. (B), (C) and (D) only
3. (A), (C) and (D) only
4. (A) and (B) only

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Match List-I with List-II

List-I

List-II

\[\text{Differential Equation} \]

Order and Degree

(A) $\left(\frac{d^2y}{dx^2}\right)^2 - e^x \left(\frac{dy}{dx}\right)^4 + 1 = 0$

(I) order = 1 and degree = 2

(B) $\left(\frac{dy}{dx}\right)^2 + xy = 0$

(II) order = 2 and degree = 1

(C) $\left(1 + \frac{dy}{dx}\right)^{3/2} = 4\left(\frac{d^2y}{dx^2}\right)^2$

(III) order = 2 and degree = 2

(D) $\sqrt{\frac{d^2y}{dx^2} + 1} = \frac{dy}{dx}$

(IV) order = 2 and degree = 4

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)
2. (A) – (II), (B) – (IV), (C) – (I), (D) – (III)
3. (A) – (III), (B) – (I), (C) – (II), (D) – (IV)
4. (A) – (II), (B) – (I), (C) – (III), (D) – (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The area (in sq. units) of the triangle whose vertices are (0, 0), (a, 0), (0, b), is equal to

1. a^2
2. $|ab|$
3. $\frac{1}{2}|ab|$
4. b^2

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Area (in sq. units) of the region bounded by the curve $y^2 = 4x$, y -axis and the line $y = 3$ is

1. $\frac{9}{2}$
2. $\frac{9}{4}$
3. 2

4. $\frac{9}{3}$

Options

1. 1
2. 2
3. 3
4. 4

Q.10

If $x = e^t$ and $y = e^{2t}$ then $\frac{d^2y}{dx^2} =$

1. $2x$
2. $2y$
3. 2
4. $\frac{1}{2x}$

Options

1. 1
2. 2
3. 3
4. 4

Q.11

The solution set of the inequality $3x + 5y < 4$ is

1. an open half plane not containing the origin
2. an open half plane containing the origin
3. the whole xy -plane not containing the line $3x + 5y = 4$
4. a closed half plane containing the origin

Options

1. 1
2. 2
3. 3
4. 4

Q.12

$$\int_0^2 (|x| + |x - 2|) dx =$$

1. 8
2. 4
3. 2
4. $\frac{1}{2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.13

The probability distribution of a random variable X is given below:

x	0	1	2	3
$P(x)$	$\frac{k}{3}$	$\frac{k}{2}$	$\frac{k}{4}$	$\frac{k}{7}$

Then the value of k is

1. $\frac{103}{84}$

2. $\frac{84}{103}$
3. $\frac{84}{102}$
4. $\frac{84}{101}$

Options 1. 1

2. 2
3. 3
4. 4

Q.14 $\int \frac{\sin 2x \, dx}{\sqrt{9 - \cos^4 x}}$ equals

1. $\sin^{-1}\left(\frac{\cos^2 x}{3}\right) + C$: C is an arbitrary constant
2. $\cos^{-1}\left(\frac{\sin^2 x}{3}\right) + C$: C is an arbitrary constant
3. $\sin\left(\frac{\cos^2 x}{3}\right) + C$: C is an arbitrary constant
4. $-\sin^{-1}\left(\frac{\cos^2 x}{3}\right) + C$: C is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.15 If, $A = \begin{bmatrix} x & 3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ y & 3 \end{bmatrix}$ and, $C = \begin{bmatrix} z & 1 \\ 8 & 2 \end{bmatrix}$ are singular matrices then:

- (A) $x > y$
- (B) $y > z$
- (C) $z > x$
- (D) $x \neq y \neq z$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (C) only
3. (C) and (D) only
4. (D) only

Options 1. 1

2. 2
3. 3
4. 4

Section : Applied Mathematics

Q.16 Which of the following are correct about t-test statistics?

- (A) The mean of t-test distribution is 0.
- (B) It depends on the degrees of freedom.
- (C) It depends on population standard deviation.

(D) With more degrees of freedom, it closely resembles standard normal distribution.

Choose the \textbf{correct} answer from the options given below:

1. (A) and (B) only
2. (B), (C) and (D) only
3. (A), (B), (C) and (D)
4. (A), (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.17 A 95\% confidence interval for a population mean was reported to be 152 to 160, if sample standard deviation $\sigma = 15$,

then sample size used in this study is (Given $Z_{0.025} = 1.96$)

1. 60
2. 50
3. 52
4. 54

Options

1. 1
2. 2
3. 3
4. 4

Q.18 The curve $x = y^2$ and $xy = k$ cut orthogonally, then k^2 is equal to:

1. 1
2. $\frac{1}{2}$
3. $\frac{1}{4}$
4. $\frac{1}{8}$

Options

1. 1
2. 2
3. 3
4. 4

Q.19

Match **List-I** with **List-II**

List-I	List-II
Matrix Product	Order of resultant matrix
(A) $[a_{ij}]_{2 \times 3} \times [b_{ij}]_{3 \times 4}$	(I) 2×4
(B) $[a_{ij}]_{2 \times 1} \times [b_{ij}]_{1 \times 3}$	(II) Not possible
(C) $[a_{ij}]_{3 \times 2} \times [b_{ij}]_{3 \times 2}$	(III) 3×3
(D) $[a_{ij}]_{3 \times 3} \times [b_{ij}]_{3 \times 3}$	(IV) 2×3

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
- (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.20

Consider the matrices $A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 25 \end{bmatrix}$ and $B = \begin{bmatrix} 1/5 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$. The value of $|(AB)^{-1}|$ is

- $\frac{15}{212}$
- 60
- $\frac{212}{15}$
- $\frac{1}{60}$

Options 1. 1

2. 2

3. 3

4. 4

Q.21 The area bounded by $y = 3x + 1$, $x = 0$, $y = 0$ and $x = a$ is 8 Sq.units. Then value of a (where $a > 0$) is

- 1
- 2
- 3
- $8/3$

Options 1. 1

2. 2

3. 3

4. 4

Q.22

A boat covers a distance of 51 km downstream in 3 hours and takes $7\frac{2}{7}$ hours to cover the same distance upstream. If

the boat's speed in still water is 12 km/hrs, then the speed of the stream is:

1. 3 km/hr
2. 5 km/hr
3. 7 km/hr
4. 11 km/hr

Options 1. 1
2. 2
3. 3
4. 4

Q.23

Suppose X has Poisson distribution such that $3P(X = 1) = 2P(X = 2)$ then $P(X > 0)$ is:

1. $1 - e^3$
2. $\frac{2}{3}e^3 - 1$
3. $1 - e^{-3}$
4. $\frac{2}{3}e^{-3} - 1$

Options 1. 1
2. 2
3. 3
4. 4

Q.24

Match List-I with List-II

List-I	List-II
(A) Marginal average cost if cost function $C(x) = \frac{50}{\sqrt{x}}$	(I) $50\sqrt{x}$
(B) Marginal average cost if cost function $C(x) = 50\sqrt{x}$	(II) $-\frac{75}{x^2\sqrt{x}}$
(C) Revenue function if demand function $P = \frac{50}{\sqrt{x}}$	(III) $\frac{-25}{x\sqrt{x}}$
(D) Marginal revenue if demand function $P = 50\sqrt{x}$	(IV) $75\sqrt{x}$

Choose the **correct** answer from the options given below:

1. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1
2. 2
3. 3
4. 4

Q.25

The maximum value of the objective function $Z = 10x + 15y$ of an L.P.P, subjected to the constraints

$$2x + 4y \leq 8,$$

$$3x + y \leq 6,$$

$$-x - y \geq -4,$$

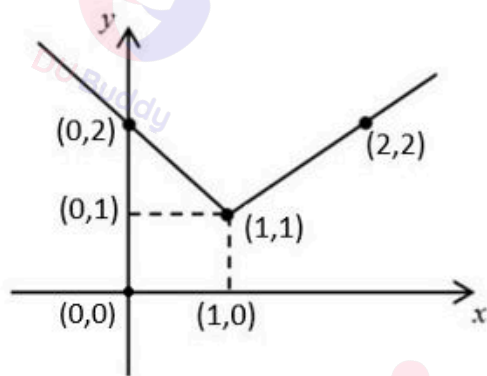
$x \geq 0, y \geq 0$ is:

1. 60
2. 38
3. 30
4. 34

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.26

The graph given below represents which of the following function?



1. $y = |x + 1|$
2. $y = |x + 1| + 1$
3. $y = |x - 1| + 1$
4. $y = |x + 1| - 1$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.27 For a Binomial distribution, $B(n, p)$, where $p + q = 1$, the sum and product of mean and variance are 8 and 12 respectively, when the value of n is:

1. 6
2. 9
3. 12
4. 16

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.28

If $\int \frac{2x-5}{(2x-3)^3} e^{2x} dx = \frac{\lambda e^{2x}}{(2x-3)^2} + C$, where C is an arbitrary constant then the value of λ is:

1. $\frac{1}{2}$
2. 2
3. -2
4. $-\frac{1}{2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.29

If three year moving averages for five data items given by 16, 18, 20, 10, 21 are x, y and z respectively, then:

1. $x > y > z$
2. $y > x > z$
3. $y < z < x$
4. $z > y > x$

Options

1. 1
2. 2
3. 3
4. 4

Q.30

Siyaaram has 100 Kg apples, he sells a part of it at 8\% profit and rest of it at 18\% profit. The overall profit he earns is 14\%. The quantity he sold at 18\% profit is:

1. 71 Kg
2. 30 Kg
3. 60 Kg
4. 55 Kg

Options

1. 1
2. 2
3. 3
4. 4

Q.31

The rate of change of the area of a circle with respect to its radius r , when $r = 3\text{cm}$, is:

1. $6\pi \text{ cm}^2/\text{cm}$
2. $4\pi \text{ cm}^2/\text{cm}$
3. $2\pi \text{ cm}^2/\text{cm}$
4. $3\pi \text{ cm}^2/\text{cm}$

Options

1. 1
2. 2
3. 3
4. 4

Q.32

Match **List-I** with **List-II**

List-I	List-II
Example	Time-series component
(A) Labour strike	(I) Secular-trend
(B) Continuous decline in death rate	(II) Seasonal
(C) Rise in prices before Diwali	(III) Cyclical
(D) Rise and fall of the share -market	(IV) Irregular

Choose the **correct** answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Rajesh calculated 95\% confidence level. What does he mean by that?

- (A) He can be 95\% confident that his sample will include the population parameter.
- (B) He can be 95\% confused that his sample will include the population parameter.
- (C) He can be 5\% confident that his sample will not include the population parameter.
- (D) He can be 5\% confident that his sample will include the population parameter.

Choose the **correct** answer from the options given below:

1. (A) and (C) only
2. (B) and (C) only
3. (D) only
4. (A) and (B) only

Options 1. 1

2. 2

3. 3

4. 4

Q.34 The values of 'a' and 'b' if the equation of straight line trend by least square method is given by $y = a + bx$ such that

$\sum x = 0, \sum y = 84, \sum xy = 108, \sum x^2 = 70$ for 6 observation as are:

1. $a = 0; b = 1.28$
2. $a = 14; b = 1.54$

3. $a = 11.6; b = 0$
4. $a = 14; b = 1.28$

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Which of the following statements are true?

- (A) The function $f(x) = \frac{x^4}{4} - \frac{4}{3}x^3 + \frac{x^2}{2} + 6x$ has 3 critical points.
(B) The function $f(x) = |x| + 3$ has no minimum value.
(C) A local maximum value is always the absolute maximum value.
(D) $f(x) = x^2$ has minima at $x = 0$.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (C) only
3. (A) and (D) only
4. (A), (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Question:

Anush takes a loan of ₹ 150000 @ 16% annual interest for 5 years. His EMI

(Equally Monthly Installment) on monthly basis under flat rate system is:

1. ₹ 4500
2. ₹ 5000
3. ₹ 4800
4. ₹ 5400

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Question:

The random variable X can take values 0, 1, 2. If $P(X = 0) = P(X = 1) = \alpha$, and $E(X^2) = E(X)$,

then which of the following are correct?

- (A) $E(X) = 2 - 3\alpha$
(B) $E(X^2) = 4 + 7\alpha$

(C) $\alpha = \frac{1}{2}$

(D) $\alpha = \frac{1}{5}$

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (C) and (D) only
3. (B) and (C) only
4. (A) and (C) only

Options

1. 1
2. 2
3. 3
4. 4

Q.38 Two pipes P and Q can fill a tank in 26 minutes and 52 minutes respectively.

Both the pipes are opened together for some time and then pipe P is closed.

If the tank is filled in 26 minutes, then after how many minutes pipe P is closed?

1. 19 minutes
2. 20 minutes
3. 13 minutes
4. 10 minutes

Options

1. 1
2. 2
3. 3
4. 4

Q.39 A machine costing ₹ 36000 has an effective life of 5 years with scrap value

of ₹ 5000 following a linear method of depreciation. Which are correct?

- (A) The value of the machine after 1 year is ₹ 31000
- (B) The value of the machine after 2 years is ₹ 23600
- (C) The value of the machine after 3 years is ₹ 18400
- (D) The value of the machine after 4 years is ₹ 11200

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (B) and (C) only
3. (B) and (D) only
4. (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.40 Question:

The unit's digit of 13^{13} is:

1. 2
2. 5
3. 3
4. 9

Options

1. 1
2. 2
3. 3
4. 4

Q.41 Question:

The value of $\begin{vmatrix} 7! & 8! & 9! \\ 8! & 9! & 10! \\ 9! & 10! & 11! \end{vmatrix}$ is :

1. $2 \times 9! \times 10! \times 11!$
2. $2 \times 8! \times 9! \times 10!$
3. $2 \times 7! \times 8! \times 9!$
4. $2 \times 10! \times 11!$

Options

1. 1
2. 2
3. 3
4. 4

Q.42 The declared rate of return compounded semi annually equivalent to the effective rate of return 10.25\% per annum is:

1. 10.13\% p.a.
2. 9.89\% p.a.
3. 10\% p.a.
4. 10.05\% p.a.

Options

1. 1
2. 2
3. 3
4. 4

Q.43 Question:

If $\begin{bmatrix} ab & cd \\ a+c & b+d \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix}$ where a, b, c, d are integers, then which of the following are true?

- (A) $a + d = 0$
- (B) $b + d = 3$
- (C) $b + d = 1$
- (D) $c + d = 2$

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (C) and (D) only
3. (B) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Question:

A die is thrown 4 times and getting 3 is considered a success. The probability

of 2 successes is :

1. $\frac{25}{1296}$
2. $\frac{1250}{72}$
3. $\frac{25}{216}$
4. $\frac{25}{72}$

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Question:

If it is 7:00 pm currently in the clock, what will the clock show

(in am or pm) after 674 hours?

1. 9:00 am
2. 2:00 pm
3. 9:00 am
4. 9:00 pm

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Rahul invested ₹ 20000 in a mutual fund in year 2018. If the value of mutual

fund increased to ₹ 32000 in year 2023, then the compound annual growth

rate of his investment is: $\left[\text{given that } (1.6)^{\frac{1}{5}} = 1.098 \right]$

1. 10.5\%
2. 9.4\%
3. 9.8\%
4. 8.8\%

Options 1. 1

2. 2

3. 3

4. 4

Q.47

The solution of the system of equations $2x + \frac{1}{2}y - z = 1$, $2y = 3$, $x + 2z = 4$ is :

1. $x = 0.5$, $y = 1.5$, $z = 1.5$

2. $x = 0.9$, $y = 1.5$, $z = 1.5$

3. $x = 1.5$, $y = 1.55$, $z = 0.9$

4. $x = 0.9$, $y = 1.5$, $z = 1.55$

Options 1. 1

2. 2

3. 3

4. 4

Q.48

Ajesh has set up a sinking fund in order to have ₹ 10,00,000 after 10 years

for his son's education. The amount should be set aside at the end of every

6 months into an account paying 5\% per annum compounded half yearly is:

[given $(1.025)^{20} = 1.6386$]

1. ₹ 37438.8

2. ₹ 3,091.48

3. ₹ 39,148.1

4. ₹ 3914.00

Options 1. 1

2. 2

3. 3

4. 4

Q.49

At what rate of interest will the present value of a perpetuity of ₹ 500

payable at the end of every 6 months be ₹ 20,000?

1. 10\% per annum

2. 20\% per annum

3. 5\% per annum

4. 7\% per annum

Options 1. 1

2. 2

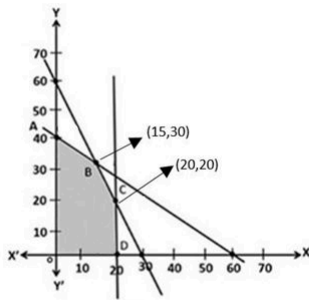
3. 3

4. 4

Q.50

From the below-mentioned graph of shaded feasible region of a linear programming problem (LPP) with objective function

$z = 1.50x + 1.00y$, the maximum value of z will be:-



1. 50
2. 52.5
3. 61.3
4. 72.4

Options 1. 1

2. 2

3. 3

4. 4

Section : Core Mathematics

Q.51 Let $\vec{a}$ be a non-zero vector of magnitude a and λ is a non-zero scalar, then $\lambda\vec{a}$ is a unit vector if

1. $\lambda = -1$

2. $a = \frac{1}{|\lambda|}$

3. $\lambda = 1$

4. $a = |\lambda|$

Options 1. 1

2. 2

3. 3

4. 4

Q.52 For any vector $\vec{a}$, the value of $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$ is equal to:

1. $2|\vec{a}|^2$

2. $3|\vec{a}|^2$

3. $4|\vec{a}|^2$

4. $|\vec{a}|^2$

Options 1. 1

2. 2

3. 3

4. 4

Q.53 If $P \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix}$, then matrix P is equal to:

1. $\begin{bmatrix} 1 & 2 \\ -2 & 0 \end{bmatrix}$

2. $\begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 0 \\ 2 & -2 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.54 Match List-I with List-II

List-I

(A) A square matrix P is said to be non-singular if

(B) A square matrix P is said to be singular if

(C) If a matrix P is both symmetric and skew-symmetric, then (III) $|P| \neq 0$

(D) If P is a square matrix, then

List-II

(I) $|P| = 0$

(II) PP^T is symmetric

(IV) P is a null matrix

Choose the correct answer from the options given below:

1. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)

2. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)

3. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)

4. (A) – (I), (B) – (IV), (C) – (II), (D) – (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.55 If the minimum value of the objective function $Z = ax + by$ of an LPP occurs at two points (3, 5) and (5, 3), then

1. $b = 2a$

2. $a = 2b$

3. $a = b$

4. $a = -2b$

Options 1. 1

2. 2

3. 3

4. 4

Q.56 Match List-I with List-II

List-I :

(Equation of Lines)

(A)

$$\vec{r} = (3\hat{i} - 2\hat{j} + \hat{k}) + \lambda(\hat{j} - 2\hat{k})$$

(B)

$$\frac{2-x}{1} = \frac{2y+1}{4}, z=2$$

List-II:

(Information)

(I)

Direction ratios are 2, 4, -1

(II)

Perpendicular to $2\hat{i} - \hat{j} + \hat{k}$

(C) $\frac{x}{1} = \frac{y-3}{2} = \frac{3-4z}{2}$

(III) Passing through the point $(3, -2, 1)$

(D) $\vec{r} = (3\hat{i} + 2\hat{j} + \hat{k}) + \lambda(2\hat{i} + \hat{j} - 3\hat{k})$ Direction ratios are $-1, 2, 0$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
2. (A) – (I), (B) – (IV), (C) – (II), (D) – (III)
3. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)
4. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.57 If the function $f(x) = 2x^2 - kx + 7$ is increasing on $[1, 2]$, then k lies in the interval

1. $(-\infty, 8)$
2. $(-\infty, 4)$
3. $(4, \infty)$
4. $(8, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.58 The area of the smaller region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the straight line $3x + 4y = 12$ is:

1. $(6\pi - 3)$ sq units
2. $(3\pi + 6)$ sq units
3. $(6\pi + 3)$ sq units
4. $(3\pi - 6)$ sq units

Options 1. 1

2. 2
3. 3
4. 4

Q.59 Consider the LPP: Maximize $z = 5x + 3y$ subject to $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x, y \geq 0$.

The optimal feasible solution occurs at

1. $(2, 0)$ only
2. neither $(2, 0)$ nor $\left(\frac{20}{19}, \frac{45}{19}\right)$
3. $(0, 3)$ only
4. $\left(\frac{20}{19}, \frac{45}{19}\right)$ only

Options 1. 1

2. 2

3. 3

4. 4

Q.60 If A and B are symmetric matrices of order 3×3 then the matrix $2AB - BA$ is:

1. a symmetric matrix

2. a skew-symmetric matrix

3. both symmetric and skew-symmetric matrix

4. neither symmetric nor skew-symmetric matrix

Options 1. 1

2. 2

3. 3

4. 4

Q.61 Area of the bounded region between the curve $y = |x - 2|$ and the line $y = 2$ is:

1. 2 square units

2. 4 square units

3. 6 square units

4. 8 square units

Options 1. 1

2. 2

3. 3

4. 4

Q.62 Acute angle between the lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$ and $\frac{x-1}{4} = \frac{y+1}{-3} = \frac{z+10}{5}$ is:

1. $\frac{\pi}{2}$

2. $\frac{\pi}{3}$

3. $\frac{\pi}{6}$

4. $\frac{\pi}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.63 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$ then

1. f is one-one but not onto.

2. f is onto but not one-one

3. f is both one-one and onto

4. f is neither one-one nor onto

Options 1. 1

2. 2

3. 3

4. 4

Q.64

If A^{-1} exists for the matrix $A = \begin{bmatrix} 1 & \lambda & -1 \\ -1 & 1 & 0 \\ \lambda & 1 & 1 \end{bmatrix}$ then

1. $\lambda = -1$
2. $\lambda \neq -1$
3. $\lambda \neq 2$
4. $\lambda \neq 1$

Options 1. 1

2. 2
3. 3
4. 4

Q.65

A relation R on the set $A = \{1, 2, 3, \dots, 13, 14\}$ defined as $R = \{(x, y) : 3x - y = 0\}$ is

1. Reflexive and symmetric but not transitive
2. Neither reflexive nor transitive but symmetric
3. Neither symmetric nor transitive but reflexive
4. Neither reflexive nor symmetric nor transitive

Options 1. 1

2. 2
3. 3
4. 4

Q.66

Match List-I with List-II (Where $\mathbb{R}$ is the set of real numbers)

List-I

List-II

- | | |
|---|--|
| (A) $\sin x$ is continuous on: | (I) $\mathbb{R} - \{0\}$ |
| (B) $\tan x$ is continuous on: | (II) $\mathbb{R}$ |
| (C) $\cot x$ is continuous on: | (III) $\mathbb{R} - \{n\pi : n \in \mathbb{Z}\}$ |
| (D) x^{-n} , $n \in \mathbb{N}$ is continuous on: | (IV) $\mathbb{R} - \{(2n+1)\frac{\pi}{2} : n \in \mathbb{Z}\}$ |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (II), (B) - (III), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.67

Bag I contains 3 black and 2 white balls, Bag II contains 2 black and 4 white balls. A bag is selected at random

and then a ball is drawn from it. The probability that the ball drawn is black is:

1. $\frac{7}{15}$
2. $\frac{11}{15}$

3. $\frac{14}{15}$
4. $\frac{13}{15}$

Options 1. 1
2. 2
3. 3
4. 4

Q.68

The particular solution of the differential equation $\{x \sin^2(\frac{y}{x}) - y\} dx + x dy = 0$, $y = \frac{\pi}{4}$ when $x = 1$ is

1. $x = e^{\tan^2(\frac{y}{x})+1}$
2. $x = e^{\tan^2(\frac{y}{x})-1}$
3. $x = e^{\cot(\frac{y}{x})-1}$
4. $x = e^{\cot(\frac{y}{x})+1}$

Options 1. 1
2. 2
3. 3
4. 4

Q.69

The value of $\cot\left(\cos^{-1}\frac{7}{25}\right)$ is

1. $\frac{25}{24}$
2. $\frac{24}{25}$
3. $\frac{7}{25}$
4. $\frac{7}{24}$

Options 1. 1
2. 2
3. 3
4. 4

Q.70

A boat 10 m high floating at a uniform speed of 13 meters per minute (m/min) away from a lamp post 15 m high.

Then the rate at which the length of shadow of the boat increases is:

1. 26 m/min
2. 13 m/min
3. 52 m/min
4. 6.5 m/min

Options 1. 1
2. 2
3. 3
4. 4

Q.71

If $\int_0^1 \frac{e^x}{1+x} dx = m$, then the value of $\int_0^1 \frac{e^x}{(1+x)^2} dx$ is:

1. $m - e + 1$
2. $\frac{m}{2} + e - 1$

3. $m - \frac{e}{2} + 1$

4. $m + \frac{e}{2} - 1$

Options 1. 1

2. 2

3. 3

4. 4

Q.72

If A and B are independent events and $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ then

Match List-I with List-II

List-I

List-II

(A) $P(A \cap B)$

(I) $\frac{1}{2}$

(B) $P(\bar{A})P(B) + P(A)P(\bar{B})$ (II) $\frac{1}{3}$

(C) $P(A | B) + P(B | A)$ (III) $\frac{1}{6}$

(D) $P(\bar{A} \cap \bar{B})$ (IV) $\frac{5}{6}$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)

2. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

3. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)

4. (A) – (III), (B) – (II), (C) – (I), (D) – (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.73

If the function

$$f(x) = \begin{cases} ax + 2, & x \leq 1, \\ x^2 + 3x + b, & x > 1 \end{cases}$$

is differentiable at $x = 1$, then the value of $(2a + b)$ is

1. 13

2. 11

3. 16

4. 8

Options 1. 1

2. 2

3. 3

4. 4

Q.74

If $\vec{a} = 2\hat{i} + m\hat{j} - n\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 4\hat{k}$

such that $\vec{a} = 2\vec{b}$, then the value of $14l + m + n$ is:

1. 0

2. 1
3. 2
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.75

$$\int \frac{dx}{x^3 \sqrt{1+x^4}} =$$

1. $\frac{1}{4x^2} \sqrt{1+x^4} + C$, where C is an arbitrary constant
2. $\frac{1}{x^2} \sqrt{1+x^4} + C$, where C is an arbitrary constant
3. $\frac{1}{2x^2} \sqrt{1+x^4} + C$, where C is an arbitrary constant
4. $-\frac{1}{2x^2} \sqrt{1+x^4} + C$, where C is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.76

The co-ordinates of the point where the line

$$\frac{x+3}{3} = \frac{y-1}{-1} = \frac{z-5}{-5}$$

cuts the yz -plane are:

1. (1, 0, 0)
2. (0, 0, 0)
3. (6, 0, 1)
4. (0, 2, 0)

Options 1. 1

2. 2
3. 3
4. 4

Q.77

If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$, and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$, then

1. $\vec{b} = \pm(\vec{a} \times \vec{c})$
2. $\vec{a} = \pm(\vec{b} \times \vec{c})$
3. $\vec{a} = \pm 2(\vec{b} \times \vec{c})$
4. $\vec{c} = \pm 2(\vec{a} \times \vec{b})$

Options 1. 1

2. 2
3. 3
4. 4

Q.78

Match List-I with List-II

List-I**List-II**

Differential Equation Integrating Factor

(A) $\frac{dy}{dx} + 2xy = 1$ (I) x

(B) $x \frac{dy}{dx} + 2xy = 1$ (II) e^{2x}

(C) $x \frac{dy}{dx} + y = 1$ (III) x^2

(D) $x \frac{dy}{dx} + 2y = 2$ (IV) e^{x^2}

Choose the correct answer from the options given below:

1. (A) – (IV), (B) – (II), (C) – (III), (D) – (I)
2. (A) – (IV), (B) – (II), (C) – (I), (D) – (III)
3. (A) – (II), (B) – (I), (C) – (IV), (D) – (III)
4. (A) – (II), (B) – (III), (C) – (IV), (D) – (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.79 Match List-I with List-II (where c is an arbitrary constant)**List-I****List-II**

(A) $\int \tan x \, dx$ (I) $\log |\sec x + \tan x| + c$

(B) $\int \cot x \, dx$ (II) $\log |\sec x| + c$

(C) $\int \sec x \, dx$ (III) $\log |\sin x| + c$

(D) $\int \operatorname{cosec} x \, dx$ (IV) $\log |\operatorname{cosec} x - \cot x| + c$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
2. (A) – (III), (B) – (II), (C) – (I), (D) – (IV)
3. (A) – (II), (B) – (III), (C) – (I), (D) – (IV)
4. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.80If $y = \log_e \left(\frac{e^2}{x^2} \right)$ for $x \neq 0$, then $\frac{d^2y}{dx^2}$ equals

1. $-\frac{1}{x}$
2. $-\frac{1}{x^2}$
3. $\frac{2}{x^2}$
4. $-\frac{2}{x^2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.81 Let A and B be 3×3 matrices such that $A \neq B$. If $A^3 = B^3$ and $A^2B = B^2A$, then the determinant of $A^2 + B^2$ is:

1. 1
2. 4
3. 0
4. -2

Options

1. 1
2. 2
3. 3
4. 4

Q.82 If $P(A) = \frac{3}{10}$, $P(B) = \frac{2}{5}$ and $P(A \cup B) = \frac{3}{5}$, then the value of $P(B|A) + P(A|B)$ is:

1. $\frac{3}{4}$
2. $\frac{2}{3}$
3. $\frac{1}{6}$
4. $\frac{7}{12}$

Options

1. 1
2. 2
3. 3
4. 4

Q.83 If a , b and c are distinct prime numbers then the value of

$$\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

is equal to

1. 0
2. 1
3. $(a-b)(b-c)(c-a)$
4. $(a+b)(b+c)(c+a)$

Options

1. 1
2. 2
3. 3
4. 4

Q.84

The maximum value of the determinant of the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin x & 1 \\ 1 + \cos x & 1 & 1 \end{bmatrix}$$

(where x is real) is:

1. $\sqrt{2}$
2. $\frac{1}{2}$
3. $\frac{\sqrt{3}}{2}$
4. $\sqrt{3}$

Options

1. 1
2. 2
3. 3
4. 4

Q.85 Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards.

The probability distribution of number of aces is given by:

1.

$$\begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(x) & \frac{24}{169} & \frac{1}{169} & \frac{144}{169} \end{array}$$

2.

$$\begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(x) & \frac{24}{169} & \frac{144}{169} & \frac{1}{169} \end{array}$$

3.

$$\begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(x) & \frac{144}{169} & \frac{24}{169} & \frac{1}{169} \end{array}$$

4.

$$\begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(x) & \frac{1}{169} & \frac{144}{169} & \frac{24}{169} \end{array}$$

Options

1. 1
2. 2
3. 3
4. 4